



ECO4108Z - FUTURES, OPTIONS & DERIVATIVES (2025)

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Course Overview

This course exposes students to derivative instruments including futures, options, and swaps, with applications in risk management and trading. Focus is placed on practical usage in commodity, credit, and equity markets, supported by theoretical foundations such as option strategies and their sensitivities (the Greeks) and hedging strategies using futures.

Lecture Schedule

Lecture Date	Lecture Topic
Tuesday, 05 Aug 17h30 – 20h30	Overview of course and assignment, and recap of Derivatives' basics
Thursday, 07 Aug 17h30 - 20h30 Tuesday, 12 Aug 17h30 – 20h30	Credit Derivatives - Credit Risk and The Structuring of Cash Flows - Balance Sheet and Arbitrage Collateralized Debt Obligations (CDOs) - Credit Default Swaps - Griffin and Tang (2021) <i>Did Subjectivity Play a Role in CDO Credit Ratings?</i> - Cornaggia et al., (2016) <i>Revolving doors on Wall Street</i> - Casey (2025) <i>Silicon Valley Bank: A Case Study in Post-Crisis Bank Failures</i>
Thursday, 14 Aug 17h30 - 20h30 Tuesday, 19 Aug 17h30 - 20h30 Thursday, 21 Aug 17h30 - 20h30	Equity Derivatives - Option Trading Strategies - Option Valuation - Option Sensitivities (the Greeks)



Thursday, 28 Aug 17h30 - 20h30 Tuesday, 02 Sep 17h30 - 20h30 Thursday, 04 Sep 17h30 - 20h30	Commodity Derivatives - Hedging with Futures - Spot Futures Parity - Case Studies: Derivative Mishaps
Tuesday, 26 Aug 2025 17h30 – 20h30	Derivatives Class Test (20%)
Saturday, 6 Sep 2025 09h00 - 12h00	Assignment presentations and executive summaries due (20%)
Wednesday, 11 Sep 17h30 – 20h30	Exam Scope and Revision
Lectures will take place in ECO LT I will be available in ECO LT an hour before lectures begin for consultation.	

Assessment Breakdown

- Test : 26 Aug 2025: 20%
- Group Assignment (Presentation + Report), 6 Sep 2025: 20%
- Final Exam 15 September 2025: 60%



Assignment Brief (20%)

Title: **Hedging Commodity Price Risk for a JSE Listed Firm**

Group Size: 8 students (please email me the names of your group by 18 Aug 2025)

Objective

To analyse the impact of commodity price risk on a JSE listed firm and recommend a viable hedging strategy using futures. Students will present a professional pitch and submit an executive summary report.

Instructions

1. Select a Firm: Choose a JSE listed company whose operations are materially exposed to commodity price risk.
2. Macroeconomic Exposure: Analyse the firm's exposure to commodity markets and macroeconomic variables that affect this exposure. Identify the main commodity of concern.
3. Hedging Strategy: Recommend a hedging strategy using futures contracts. Include a rationale, possible risks, and an assessment of how the strategy supports operational continuity.
4. Futures Contract Selection: Identify a specific futures contract and the exchange where it is traded. Justify your selection in terms of liquidity, standardisation, and contract specs.
5. Written Executive Report: A 4,000-word (max) executive summary suitable for presentation to shareholders. The report must be clear, concise, and well-motivated. This must be uploaded to Amathuba by Friday, 05 Sep 23h59.
6. Presentation (15 mins): Prepare a pitch to shareholders explaining your findings, analysis, and recommendations and be prepared for questions.

Assessment Criteria (Presentation & Report Total = 20%)

Component	Description	Marks
Macroeconomic Exposure	Clarity and depth of analysis; use of data	5
Hedging Strategy	Coherence, effectiveness, risk insight	5
Futures Contract & Exchange	Appropriateness and justification	5
Professionalism of Presentation & Report	Clarity, structure, persuasiveness	5

Amathuba

The learning platform for this course is Amathuba. While teaching will be conducted face-to-face, Amathuba will primarily be used for sharing resources and sending announcements. You will also be required to upload your executive summary submissions via Amathuba.

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